DATALAB ANALYTICS FINAL PROJECT

Predictive Models for Mental Health Risk Assessment and Classification

Student: **Opeyemi Precious Alalade**
Supervisor: **Oyeleke Olayemi**

BACKGROUND

- Mental strain and burnout are often noticed too late

- Workload, chronic stress and poor work-life balance make early signs easy to miss.

- Objective: build and compare machine learning models that classify a person as Healthy, Needs Attention or Critical, and show the result in a simple web app.

- This tool only supports screening and early action. It is not a diagnosis.

50,000	synthetic records
3	classes to predict
35	input features

OBJECTIVES

- Four metrics judge the models, because accuracy alone can mislead the model .

Metric	What it means	Why we use it
Accuracy	Share of all people classified correctly	Easy headline, but half the people are Healthy
Macro F1-score	Average F1 over the three classes, equal weight	The smaller classes count as much as the big one
Recall (Critical class)	Share of truly Critical people the model finds	Missing a Critical person is the costliest mistake
ROC-AUC	How well probabilities rank a class above the rest (0.5 = chance, 1 = perfect)	Quality that does not depend on a cut-off

DATA PREPARATION

- The data was clean, but four columns had to be removed to prevent target leakage(the column that gives away the answer).

Check	Result
Duplicate rows	0
Negative values	0
Outliers (IQR rule)	0
Columns with blanks	16 (0.5% to 2.9%)
Blanks filled with	training-set median

COLUMNS REMOVED TO PREVENT TARGET LEAKAGE

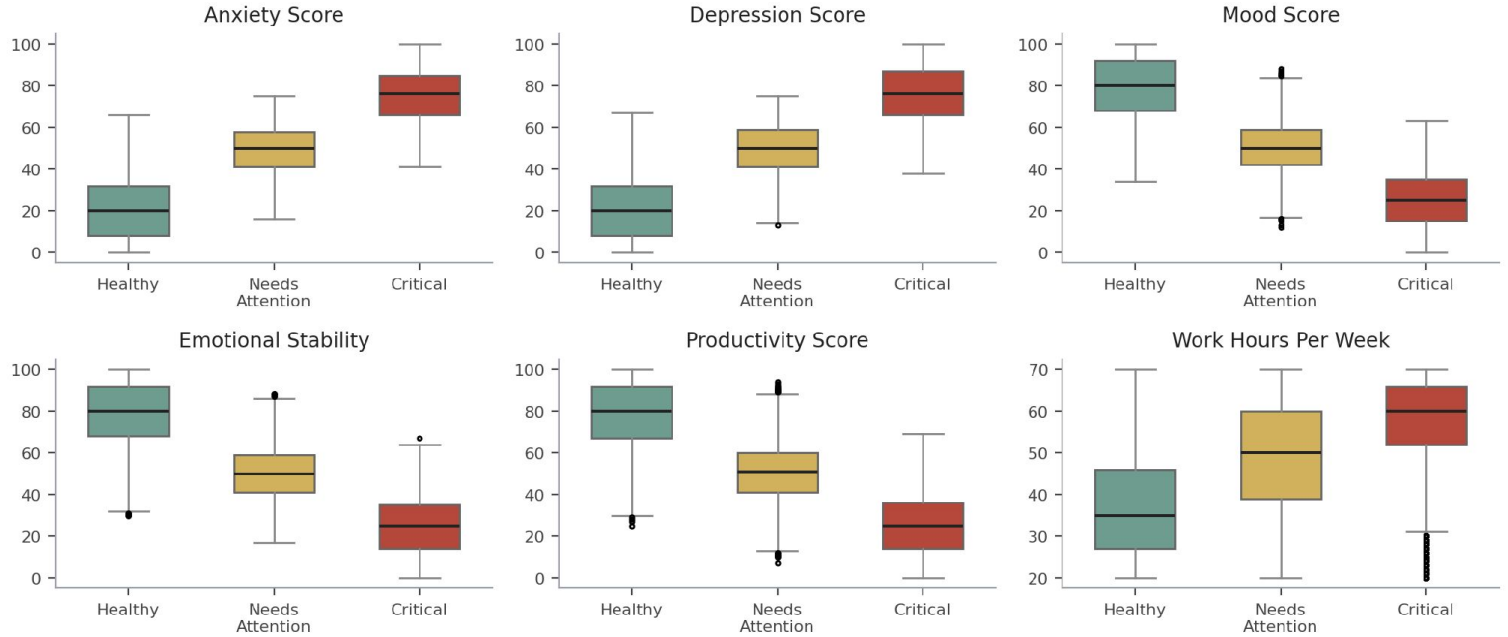
Burnout_Risk: the reveals the target in other words

Burnout_Score: the three classes are bands of this score

AI_Wellness_Recommendation: made after risk is judged, no signal

Person_ID identifier/label only:

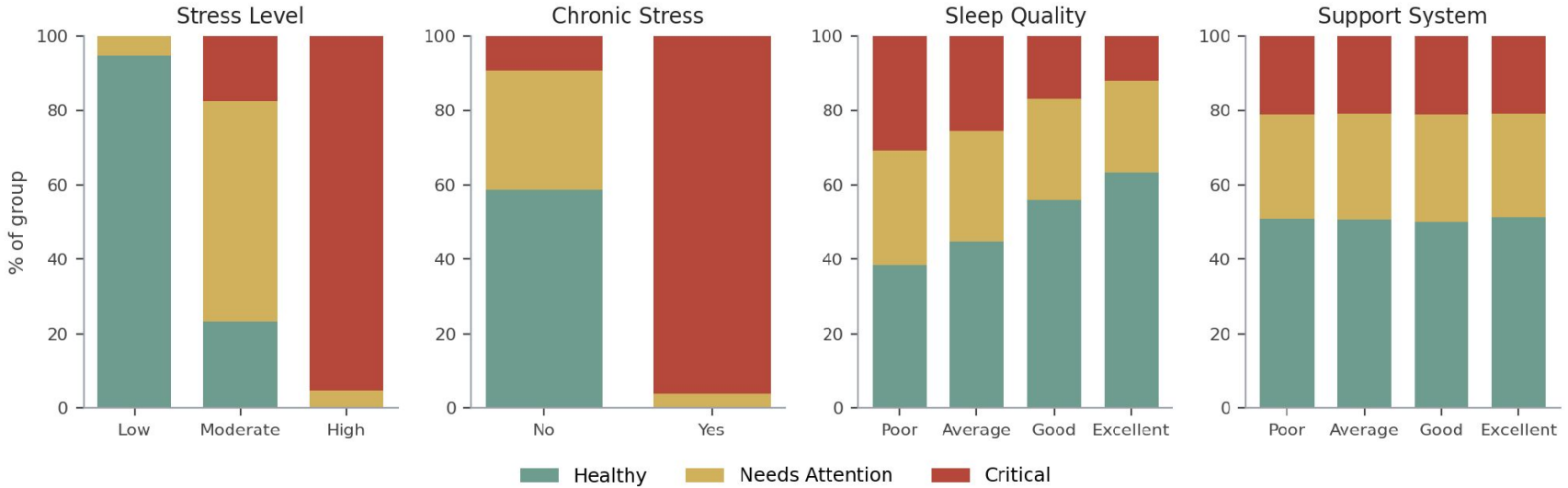
EXPLORATORY ANALYSIS



- **Box-plot Chart:** This shows that the psychological scores (Anxiety, Depression, Mood and Emotional Stability separate the three groups clearly and they are the strongest predicators for the model.
- The boxes barely overlap between Healthy and Critical, so these scores are strong clues.

EXPLORATORY ANALYSIS

- Stress appears to be the key lever for intervention and model prediction; support system gives little to nothing.

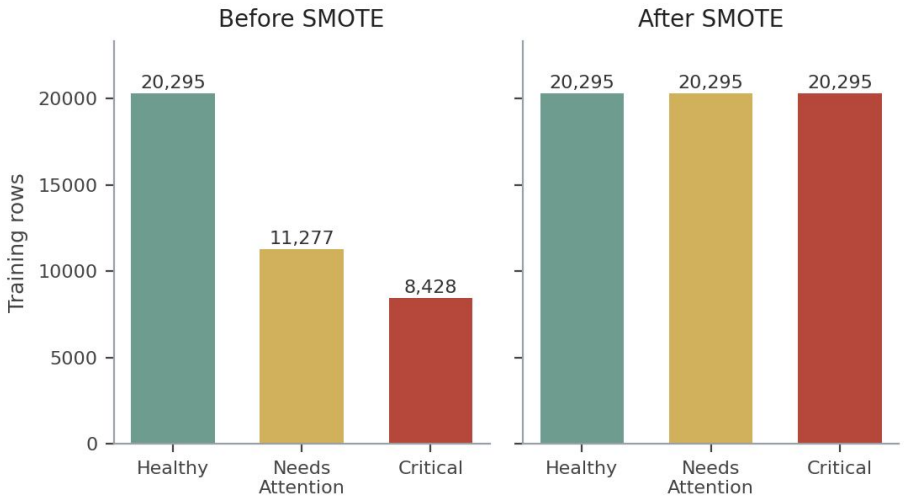


- 95% of High-stress people are Critical while roughly 95% of Low-Stress people are Healthy.
- In people that have chronic stress, about 96% arev Critical while 0% are Healthy.

DATA PREPARATION

- The data was encoded, split, scaled and balanced without leaking the test set. The steps are:

1. Encode text as numbers (36 to 86 columns)
2. Split 80/20, stratified (40,000 / 10,000)
3. Fill blanks from the training median
4. Scale with StandardScaler, fitted on training data
5. Balance training classes with SMOTE



- The test set is left unbalanced so the evaluation reflects real life.

FEATURE ENGINEERING AND SELECTION

- A combined distress index became the top-ranked feature

New feature	Built from	Correlation with status
Psych_Distress_Index	Anxiety, depression, reversed mood and stability	0.86
Work_Strain	Work hours x poor work-life balance	0.28
Income_Per_Work_Hour	Income / monthly working hours	-0.29
Recovery_Index	Activity + meditation + sleep	-0.26
Sleep_Deficit	Hours below 8	0.19

- Four other new features had almost no link to status ($|r| < 0.02$).

94 to 42

Features kept (95% of total importance). All 52 removed were country, occupation, gender and employment dummies.

MODEL EVALUATION

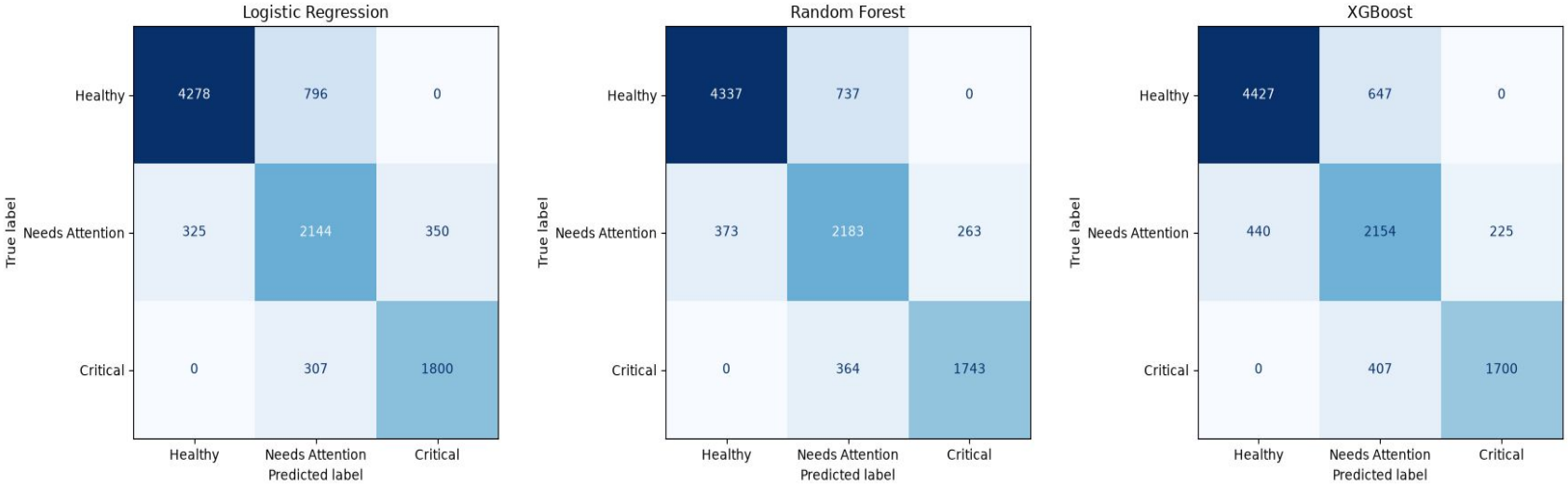
- Three models selected perform almost identically; Random Forest ranks first on macro F1.

Model	Accuracy	Macro F1	Recall (Critical)	ROC-AUC
Logistic Regression	0.8222	0.8122	0.8543	0.9499
Random Forest	**0.8263**	**0.8165**	**0.8272**	**0.9497**
XGBoost	0.8281	0.8162	0.8068	0.9509

- 5-fold cross-validation agrees: accuracy 0.822 to 0.823 for all three models, so the results are stable.
- Random Forest was selected on macro F1 (0.8165). The margin is small, and any of the three would be acceptable.

MODEL EVALUATION

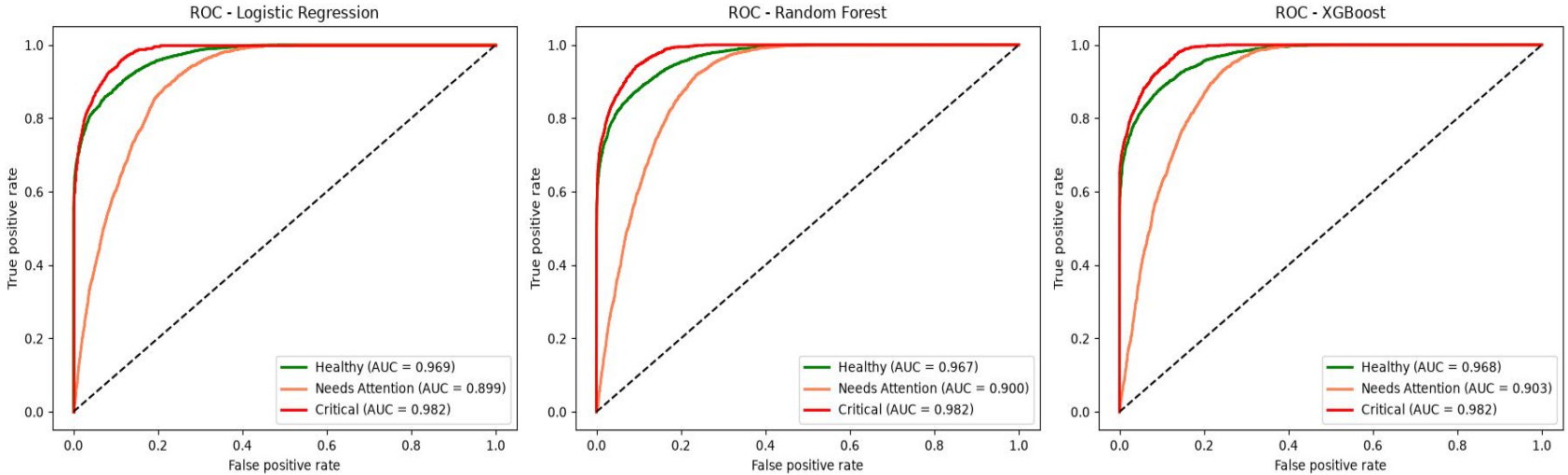
- No model ever called a Healthy person Critical, or the reverse



- Random Forest: 4,337 of 5,074 Healthy, 2,183 of 2,819 Needs Attention and 1,743 of 2,107 Critical were classified correctly. 364 Critical cases (17%) were labelled Needs Attention.

MODEL EVALUATION

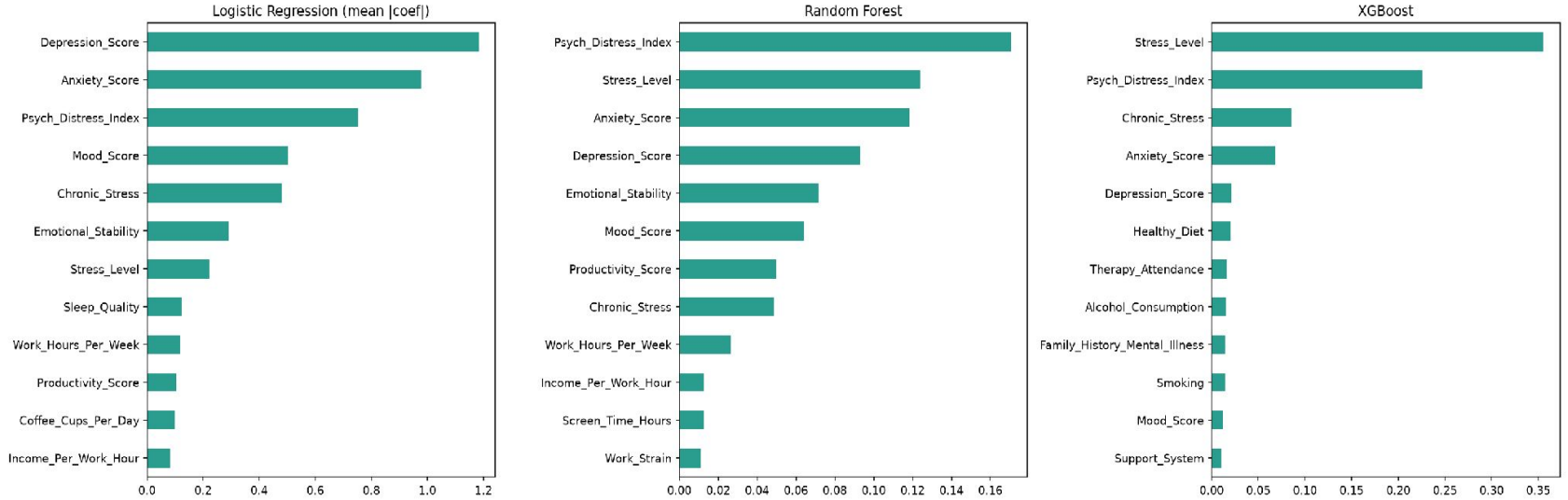
- Critical and Healthy are separated very well; Needs Attention is hardest



- One-vs-rest AUC: Critical 0.982 for all models, Healthy 0.967 to 0.969, Needs Attention 0.899 to 0.903.

MODEL EVALUATION

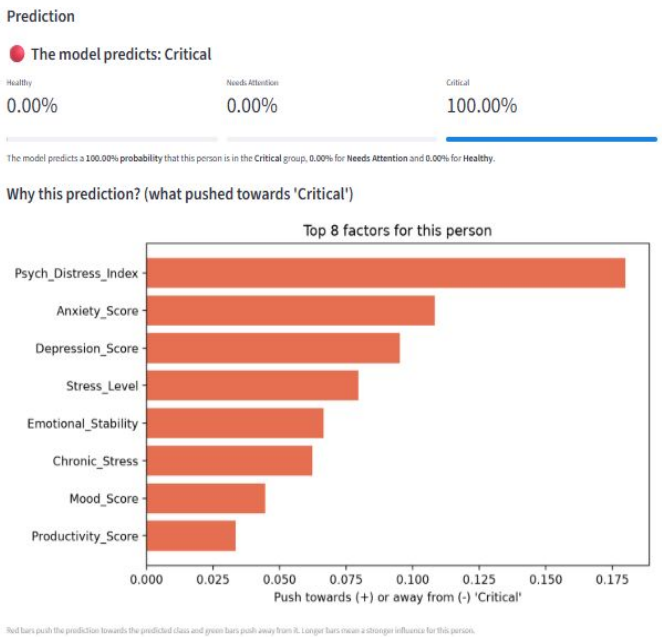
- The same top 3 factors drive every model from the graph, although they vary in ranking for each model

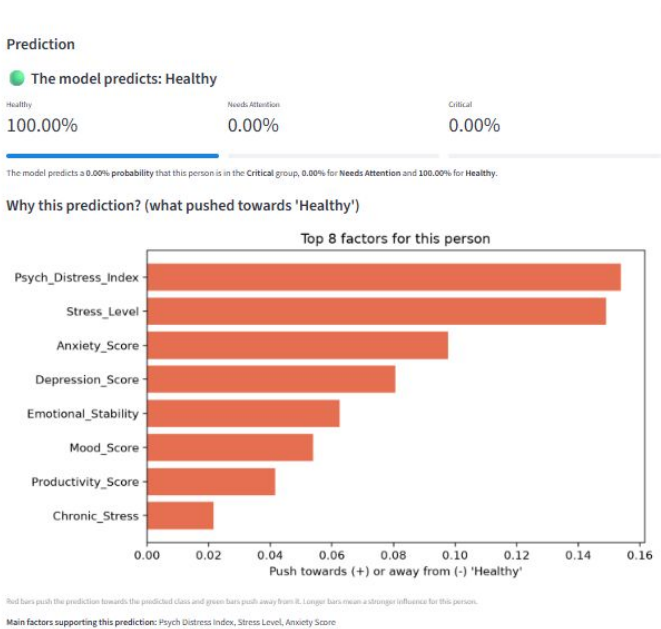


- Psych_Distress_Index is in the top three for all three models. Stress_Level carries about 36% of XGBoost importance.
- Compared bars within each chart, not across charts.

DEPLOYMENT

- A prototype app returns probabilities and explains each prediction.





Critical case: anxiety 85, depression 80, mood 20, emotional stability 20, productivity 25, High stress, chronic stress Yes

Healthy case: anxiety 15, depression 15, mood 85, emotional stability 85, productivity 85, Low stress, no chronic stress

CONCLUSION

Random Forest, selected on macro F1, reaches 82.6% accuracy, ROC-AUC 0.950 and 82.7% recall for Critical on the untouched test set. The three models are within one percentage point of each other, and no Healthy person was predicted Critical.

Recommendations

1. Use the model as a screening aid, not a diagnosis.
2. Prioritise stress management; as stress level and chronic stress are the strongest signals.
3. Address long working hours (about 58 h per week for Critical vs 37h for Healthy).
4. Support recovery: sleep, activity, meditation, less screen time and caffeine.
5. Screen on psychological and stress indicators, not on demographics.

- **Limitations:** synthetic data; Needs Attention is the hardest class.
- **Next step:** validate on real data and check fairness.